

8.

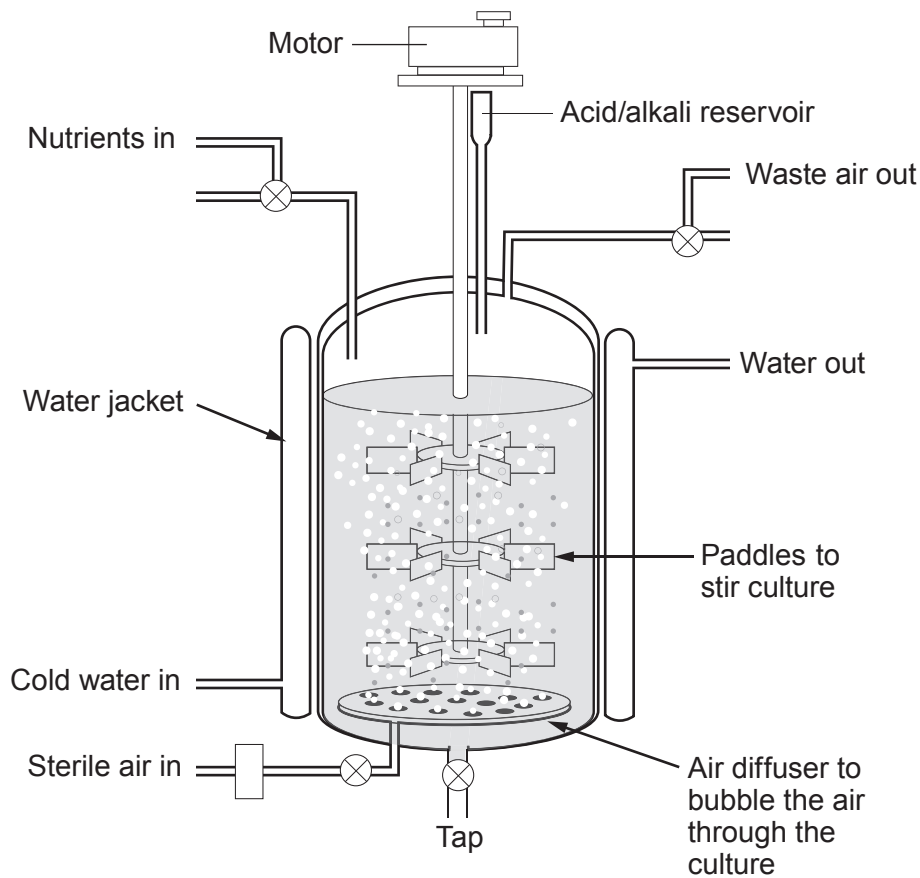
Thanks to PENICILLIN
... He will come home!



Penicillin is an antibiotic that is produced by several species of a fungus from the genus *Penicillium*. Its effect was first identified by Alexander Fleming in 1928. It was only in 1939 that Penicillin was extracted by Howard Florey and Ernst Chain at Oxford University and later developed for use as an antibiotic. The industrial scale production of penicillin was developed in the USA between 1941 and 1943 in response to the high number of soldiers who were dying due to diseases linked to their wounds during World War II.

Penicillium can be grown industrially in a tank called a fermenter. An example is shown in the diagram opposite:





(a) (i) State how penicillin helped save the lives of wounded soldiers. [1]

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(ii) Name a group of microorganisms that would not be affected by antibiotics. [1]

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(iii) State a danger of new drug treatments. [1]

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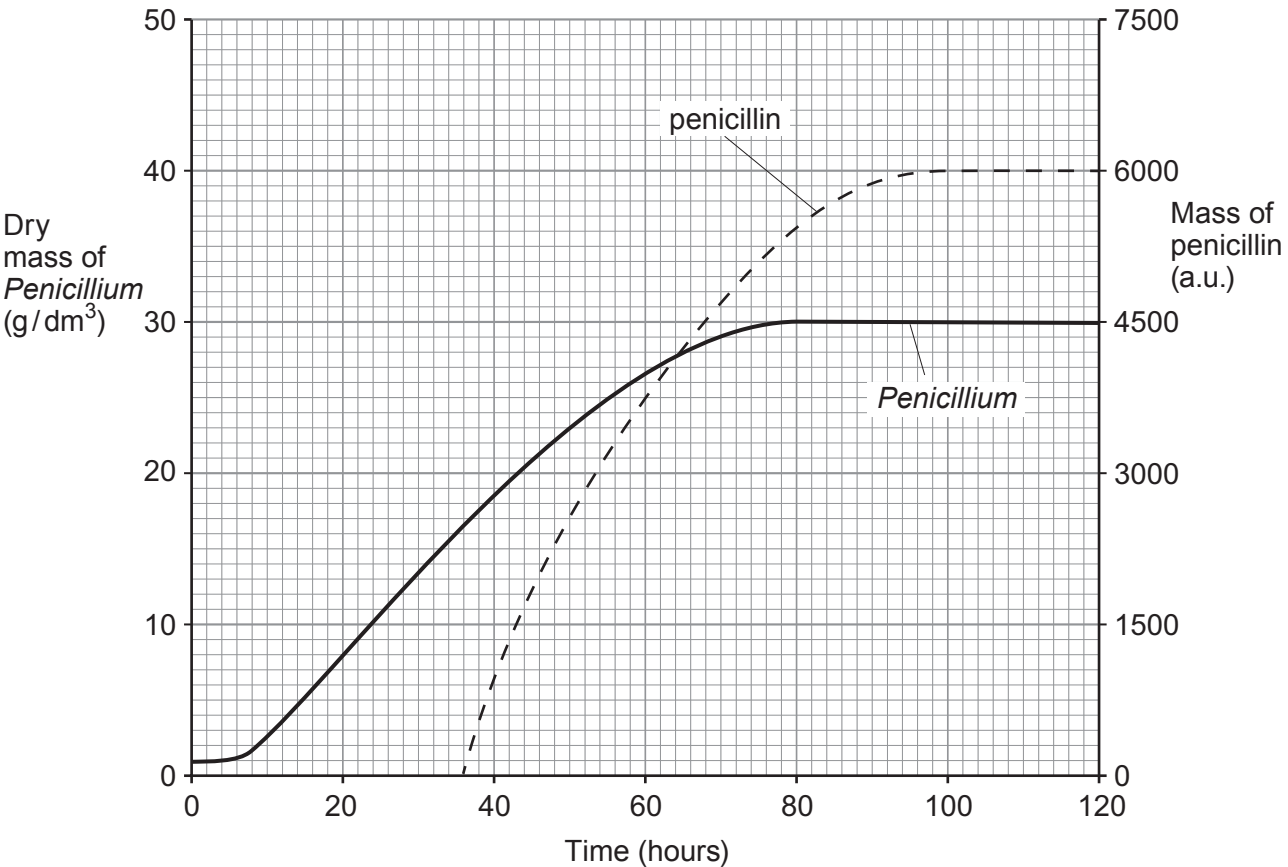
(b) Use the diagram of the fermenter and your own knowledge to answer the following question.

Other than nutrients, state **one** factor that affects the growth of *Penicillium* and state how it can be controlled by the fermenter. [1]

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(c) The graph below shows the quantity of penicillin and *Penicillium* inside the fermenter at 25°C over a period of time:



Using the graph:

- (i) Calculate the mean growth rate of *Penicillium* between 20 and 50 hours. [2]

Growth rate = g/dm³/hour



Examiner
only

(ii) Identify the optimum time for the penicillin to be extracted from the fermenter and justify your answer. [2]

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(d) Draw a line on the graph to estimate the effect of increasing temperature from 25°C to 28°C on the dry mass of *Penicillium* produced between 0-120 hours. [1]

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